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Digestive Strategies of the Wombats: Feed Intake, Fiber Digestion, and Digesta Passage in Two Grazing Marsupials with Hindgut Fermentation

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Abstract

The wombats are grazing marsupials from mesic (Vombatus ursinus) and xeric (Lasiiorhinus latifrons) regions of Australia. Feed intake and digestive responses of four wombats of each species were compared on two dietary fiber levels with pelleted straw-based diets of 59% and 44% neutral detergent fiber (NDF), and on two feed particle sizes with unpelleted and pelleted alfalfa hay. Dry matter (DM) intakes were similar between fiber levels but digestibilities of DM and NDF were greater on the low-fiber diet, with a trend to longer retention of digesta markers than on the high-fiber diet. Particulate (^{103}Ru -phenanthroline) and fluid ($^{51}\text{CrEDTA}$) digesta markers were excreted in similar patterns, but mean retention times were longer for particles than for fluids (52–62 h vs. 30–36 h on the high-fiber diet). Marker excretion patterns and digestive tract anatomy were integrated into a model of digesta flow where axial flow of digesta was combined with irregular mixing along the haustrated proximal colon. Patterns of fecal particle size distribution and DM intake of the alfalfa hays indicated a more thorough comminution of feed in the dental mill of Lasiiorhinus than of Vombatus. The digestive strategies of the wombats feature enhanced digestive tract capacity (Vombatus) or length (Lasiiorhinus) combined with low feed intakes, which result in long retention of digesta and yield fiber digestibilities similar to those of equids and macropod marsupials. These strategies enable wombats to utilize small home ranges with pastures of poor quality and/or low abundance.

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Introduction

The foraging ecology of a species is directly related to its metabolic requirements and digestive responses (see, e.g., Illius and Gordon 1992; Spalinger and Hobbs 1992). Metabolic requirements determine the level of feed intake, which is also influenced by the availability of nutrients in the diet. The digestive strategy of the species relates these dietary and metabolic constraints to parameters of feed intake, digestive tract anatomy, digesta flow, and apparent digestion (see, e.g., Baker and Hobbs 1987).

The wombats (family Vombatidae) include two species of burrowing marsupials from contrasting habitats: *Vombatus ursinus* from mesic areas of the coast and highlands and *Lasiorhinus latifrons* from semiarid regions of southern and southeastern Australia. Wombats utilize siliceous perennial grasses that are often high in fiber and low in nitrogen (N) content, especially as pastures senesce (Barboza and Hume 1992a). Rainfall and pasture growth are highly seasonal for *Lasiorhinus*, with frequent droughts that result in pastures that are often of poor quality and low quantity. Feed quality and quantity may be further constrained for both *Vombatus* and *Lasiorhinus* by small home ranges (4–12 ha; McIlroy 1976; Wells 1978) and the use of feeding territories near their burrow complexes (McIlroy 1973; Gaughwin 1981).

Plant cell contents are digested in the small, acidic stomach and the simple small intestine of the wombats. The fibrous cell walls of the grasses are digested by microbial fermentation in the hindgut. The hindgut comprises a capacious proximal colon (62%–79% of total gut contents) with only a vestigial cecum and a nonsacculated distal colon (Barboza and Hume 1992a, 1992b). Utilization of poor-quality diets is further enhanced by low maintenance requirements for energy and N in *Vombatus* and *Lasiorhinus* (Barboza, Hume, and Nolan 1993). These metabolic requirements are considerably lower than those of eutherian grazers and are the lowest reported for herbivorous marsupials.

This article describes two experiments with *Vombatus* and *Lasiorhinus* to assess their digestive strategies in response to dietary fiber level and feed particle size. Feed intake, apparent digestion, and the passage of particulate and fluid digesta markers were studied by the use of two pelleted straw-based diets of different fiber contents. The patterns of marker excretion were combined with parameters of digestive tract anatomy in a schematic model of digesta flow in the wombats. The effects of reducing dietary particle size were studied in a second experiment by monitoring feed intake, digestion, and fecal particle size distributions in wombats fed unpelleted or pelleted alfalfa hay.

Material and Methods

Maintenance procedures and experimental conditions for the wombats were described by Barboza and Hume (1992a). Each experiment included four *Vombatus* and four *Lasiobinus* in a crossover design with repeated measures for each animal on each diet. Animals were housed in metabolism cages and were allowed to adjust to experimental conditions for 4 wk. Collection periods were then commenced after feed intakes were consistent for at least 7 d. Each animal was weighed at the beginning of the collection period, during which all feed and feces were weighed daily, subsampled (20%), and stored at -20°C . At the end of the collection period, animals were reweighed, and subsamples were combined to provide bulk samples of feed refusals and feces for each animal.

Feed and feces were dried at 50°C and ground through a 1-mm screen. Dry matter (DM) content was determined by drying to constant weight at 80°C in a convection oven. Ash content was determined by combustion in a muffle furnace at 500°C for 3 h. Neutral detergent fiber (NDF) was prepared with the addition of sodium sulfite by the method of Van Soest, Robertson, and Lewis (1991).

Experiment 1: Dietary Fiber and Digesta Passage

Wombats were fed two pelleted diets of 59.1% (high fiber) and 44.1% (low fiber) NDF on a DM basis. These diets were 87.3% DM, of which 1.4% was N and 7.9% was ash. The composition (air-dry basis) of the dietary mixtures were as follows: high-fiber diet—79.0% barley (*Hordeum vulgare*) straw, 15.9% ground whole corn (*Zea mays*), and 4.0% hydrochloric casein; low-fiber diet—52.0% barley straw, 44.4% corn, and 2.5% hydrochloric casein. A mineral and vitamin supplement (Hume and Barboza 1993) was also included in both diets at 1.1% on an air-dry basis.

Each collection period of 8 d was immediately followed by a digesta passage measurement over 10 d on the same diet. The fluid phase of digesta was marked with ^{51}Cr -labeled chromium ethylenediamine tetraacetic acid (CrEDTA) (Downes and McDonald 1964; Australian Atomic Energy Commission, Lucas Heights, N.S.W.). The particulate phase of digesta was marked with ^{103}Ru -labeled tris (1,10-phenanthroline)-ruthenium (II) chloride (RuP) prepared from $^{103}\text{RuCl}_3$ (Tan, Weston, and Hogan 1971; New England Nucleonics, Sydney). Wombats were provided with feed for at least 30 min before a single oral dose of 10 mL containing 0.95 MBq RuP and 2.13 MBq CrEDTA was administered. All animals were manually restrained and lightly

anesthetized with a halothane-oxygen mixture for dosing (Barboza and Hume 1992a). Wombats recovered quickly from the dosing procedure and resumed feeding within 30 min. Excreta were collected for 235 h after dosing and stored at -20°C . Feces were collected at 4-h intervals for the first 100 h, and thereafter at intervals of 8 h, while urine was collected once each day. Times of excretion were taken as the midpoints of the collection intervals.

Markers were analyzed by gamma emission from ^{103}Ru and ^{51}Cr on dual channels in an Autogamma spectrometer (model 3002, Packard Instrument). Overlap between the channels was estimated separately for feces and urine, and counts were corrected for both overlap and background radiation. Feces were dried at 80°C in a fan-forced oven, ground in an automatic mortar and pestle, and mixed thoroughly before subsampling for counting. Feces and urine were assayed in duplicate with glass vials filled to a constant height.

Marker excretion was expressed as a proportion of the total counts per minute (cpm) excreted in both the feces and urine, that is, the recovered dose. The times for excretion of 5% (t_5), 50% (t_{50}), and 95% (t_{95}) of the marker were estimated from the cumulative excretion of marker in the feces (Warner 1981). Fecal marker concentration was expressed on a DM basis, with t_{max} as the time to maximum concentration. The mean retention time (MRT) was calculated as

$$\frac{\sum (M_i \cdot T_i)}{\sum M_i},$$

where i denotes the discrete fecal collection with marker concentration M_i at time T_i from dosing (Blaxter, Graham, and Wainman 1956; Warner 1981).

Marker specificity within the digestive tract was tested by incubating both markers with fresh digesta in vitro. Digesta were sampled from the proximal colon of four *Vombatus* and four *Lasiobrinus* fed a high-fiber diet in studies reported by Barboza and Hume (1992a, 1992b). Approximately 15 g digesta were sampled within 30 min postmortem, mixed with 23 kBq CrEDTA and 17 kBq RuP in 0.5 mL water, and incubated under anaerobic conditions at 35°C for 120 min (Barboza and Hume 1992b). Incubations were stopped by the addition of 10 mL formaldehyde solution (4% wt/vol in $9 \text{ g} \cdot \text{L}^{-1}$ NaCl) and stored at 3°C .

Fluid and particulate phases were separated by centrifugation at 20,000 G for 30 min to precipitate all particles including microbial cells. A series of four supernatants were collected from repeated rinses of centrifuged

particles with saline ($9 \text{ g} \cdot \text{L}^{-1} \text{ NaCl}$). Particles were separated into fine and coarse fractions by a $150\text{-}\mu\text{m}$ mesh (Endecotts). All fractions were transferred to glass vials for counting; particulate fractions were subsequently dried to express marker concentration on a DM basis.

Experiment 2: Pelleting the Feed and Particle Size

Pelleted and unpelleted alfalfa (*Medicago sativa*) hays were fed to *Vombatus* and *Lasiobinus* for collection periods of 7 d. A single batch of hay was chopped to approximately 10 mm length in a hammer mill, mixed thoroughly, and stored in bags. Half this batch was pelleted without the addition of water or further milling. Chemical composition was not altered by pelleting the hay; these diets were 90.2% DM, of which 38.9% was NDF, 3.2% was N, and 10.1% was ash.

Feed and feces were stored at -20°C for analysis of particle size distribution by wet sieving. Analyses were performed in triplicate with 1 g DM per assay. Samples were soaked overnight in 40 mL water and separated on a stack of six meshes (4.75 mm, 2.36 mm, 1.18 mm, $600 \mu\text{m}$, $150 \mu\text{m}$, and $75 \mu\text{m}$) mounted in 10-cm-diameter sieves (Endecotts). Sieving was performed under standardized conditions of water flow and agitation (Barboza 1989) with an apparatus similar to that of Evans et al. (1973) and Allen, Robertson, and Van Soest (1984). The contents of each sieve were transferred to filter papers and dried to express particle size distribution on a DM basis. Dry matter passing through the finest sieve was estimated by difference.

Statistics

Results were compared by ANOVA on repeated measures, with species as a factor between groups and diet as a factor within groups (Winer 1971; Tabachnick and Fidell 1983). All percentages were transformed to arcsines for statistical analysis (Zar 1974). Means are reported with the standard deviation of the sample.

Results

Experiment 1: Dietary Fiber Level and Digesta Passage

Feed intakes were similar between dietary fiber levels and between *Vombatus* and *Lasiobinus* (table 1). Digestibilities of DM and NDF were higher

TABLE 1
Intake and digestion in wombats fed high-fiber (59% NDF) and low-fiber (44% NDF) diets

Measure and Species	High Fiber	Low Fiber	Statistic
Body mass (kg):			
<i>Vombatus</i>	29.1 ± 2.7	29.8 ± 2.9	NS
<i>Lasiorbinius</i>	26.0 ± .7	26.4 ± .9	
DM intake (g · kg ^{-0.75} · d ⁻¹):			
<i>Vombatus</i>	32.7 ± 11.2	35.8 ± 7.9	NS
<i>Lasiorbinius</i>	32.6 ± 4.1	35.5 ± 3.9	
DM digestibility (%):			
<i>Vombatus</i>	43.1 ± 5.9	60.0 ± 3.1	D***
<i>Lasiorbinius</i>	41.3 ± 2.9	57.0 ± 1.8	
NDF digestibility (%):			
<i>Vombatus</i>	29.8 ± 10.2	37.5 ± 9.3	D***, I*
<i>Lasiorbinius</i>	26.7 ± 4.1	29.7 ± 6.4	

Note. Values are means ± SD for four animals of each species. Statistics are as follows: D, effect of diet; I, effect of species-by-diet interaction. Significance is for the *F* statistic; NS, not significant.
* *P* < 0.05.
*** *P* < 0.001.

on the low-fiber diet in both species. Digestibility of NDF on the low-fiber diet was higher for *Vombatus* than for *Lasiorbinius*. Retention times of digesta markers were not significantly different between fiber levels but were more variable on the low-fiber diet (table 2). There was a consistent trend toward longer retention of both markers on the low-fiber diet (table 2).

The excretion times for each marker indicated that the majority of the dose was excreted as a pulse, since *t*_{max}, *t*₅₀, and MRT were similar and approximately 17 h from both the beginning (*t*₅) and the end (*t*₉₅) of the excretion (table 2). This pulse of excretion was reflected in the marker concentrations following dosing (fig. 1), which also included minor peaks or fluctuations in the concentrations of both digesta markers at the beginning and at the end of the excretion. Although the patterns of excretion were similar for both markers, all retention times for RuP were 16–26 h longer than those for CrEDTA (table 2).

Total recoveries of marker doses in excreta were high (97% ± 5% of RuP and 99% ± 7% of CrEDTA), with only a small proportion of the recovered

TABLE 2
Retention times of fluid and particulate digesta markers in wombats fed high-fiber (59% NDF) and low-fiber (44% NDF) diets

Time and Species	High Fiber (h)	Low Fiber (h)
Fluid marker (CrEDTA):		
t_5 :		
<i>Vombatus</i>	22 ± 4	40 ± 28
<i>Lasiorbinius</i>	20 ± 2	35 ± 19
t_{50} :		
<i>Vombatus</i>	34 ± 10	45 ± 27
<i>Lasiorbinius</i>	28 ± 7	47 ± 23
$t_{\max}$:		
<i>Vombatus</i>	35 ± 9	46 ± 27
<i>Lasiorbinius</i>	28 ± 5	49 ± 25
MRT:		
<i>Vombatus</i>	36 ± 9	50 ± 29
<i>Lasiorbinius</i>	30 ± 5	49 ± 23
t_{95} :		
<i>Vombatus</i>	46 ± 6	72 ± 29
<i>Lasiorbinius</i>	48 ± 19	63 ± 25
Particulate marker (RuP):		
t_5 :		
<i>Vombatus</i>	45 ± 7	56 ± 24
<i>Lasiorbinius</i>	37 ± 6	52 ± 30
t_{50} :		
<i>Vombatus</i>	60 ± 11	72 ± 35
<i>Lasiorbinius</i>	49 ± 7	64 ± 37
$t_{\max}$:		
<i>Vombatus</i>	61 ± 12	73 ± 31
<i>Lasiorbinius</i>	49 ± 6	64 ± 36
MRT:		
<i>Vombatus</i>	62 ± 11	75 ± 35
<i>Lasiorbinius</i>	52 ± 5	69 ± 38
t_{95} :		
<i>Vombatus</i>	74 ± 13	89 ± 38
<i>Lasiorbinius</i>	62 ± 8	85 ± 46

Note. Values are means ± SD for four animals of each species. There were no significant effects of species, diet, or species-by-diet interaction ($P > 0.05$); MRT, mean retention time; t_5 , time for excretion of 5% of the marker; t_{50} , time for excretion of 50% of the marker; t_{95} , time for excretion of 95% of the marker; $t_{\max}$, time to maximum concentration.

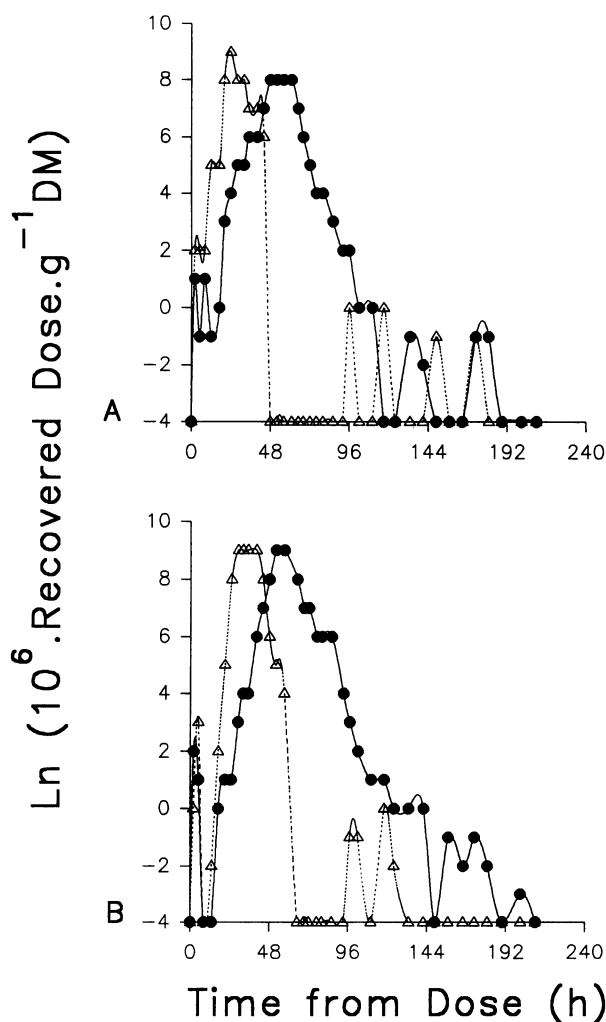


Fig. 1. Concentrations ($\ln 10^6 \cdot \text{recovered dose} \cdot \text{g}^{-1} \text{DM}$) of particulate (RuP; solid circles) and fluid (CrEDTA; open triangles) digesta markers in the feces of wombats following a single oral dose. Minima are at background counts per minute. Maxima are approximately 20,000 cpm RuP $\cdot \text{g}^{-1} \text{DM}$ and 10,000 cpm CrEDTA $\cdot \text{g}^{-1} \text{DM}$. Excretion curves are chosen randomly for one animal of each species: A, Vombatus; B, Lasiorhinus.

dose present in the urine ($0.2\% \pm 0.2\%$ RuP and $3.0\% \pm 2.2\%$ CrEDTA). This indicates that estimates of retention were not significantly affected by incomplete collection of the markers or by absorption of markers from the digestive tract (Owens and Hanson 1992).

The distribution of markers between fluid and particulate fractions of digesta indicated high phase specificity. The fluid phase contained $98.7\% \pm 0.4\%$ of the CrEDTA dose and only $2.8\% \pm 0.8\%$ of the RuP dose. The RuP was equally distributed between the fine ($49.1\% \pm 3.9\%$) and the coarse ($48.1\% \pm 4.0\%$) particulate fractions. Since the fine particles constituted only 30% of the total DM, RuP was bound to fine particles in higher concentrations than the coarse material. These results are consistent with the migration of RuP from coarse to fine particles of digesta in ruminants (Dixon, Kennelly, and Milligan 1983).

Experiment 2: Pelleting the Feed and Particle Size

Wombats did not select components of the alfalfa hays, since NDF and N contents of refused feeds were similar to those offered (*t*-test, $P > 0.05$). Dry matter intakes of both these diets were lower in *Vombatus* than in *Lasiorbatus*, with higher intakes of the pelleted hay than the unpelleted hay in *Vombatus* (table 3). Digestibilities of DM and NDF were inversely related to feed intake, with higher digestibilities for *Vombatus* than *Lasiorbatus*, especially on the unpelleted hay (table 3).

Pelleting the diet altered the distribution of particles from the largest mesh (4.75 mm) to the intermediate sizes (2.36 mm, 1.18 mm, and 600 μm) but did not increase the proportion of fine particles (150 μm or less) (fig. 2A and B). Distributions of fecal particles were more variable for the unpelleted hay than for the pelleted hay in both *Vombatus* (fig. 2C and D) and *Lasiorbatus* (fig. 2E and F). In *Vombatus*, fecal particle sizes were distributed equally among the five largest meshes for the unpelleted hay, whereas on the pelleted diet most DM was retained on the 150- μm and 600- μm meshes. Distribution patterns of fecal particles in *Lasiorbatus* were less variable than for *Vombatus*, with most DM retained on the 600- μm mesh for both diets.

Discussion

The pattern of marker excretion (fig. 1) and the digestive tract anatomy (Barboza and Hume 1992a) were combined in a schematic model of digesta flow in the wombats (fig. 3). The simple anatomy of the small intestine and the distal colon suggested a tubular flow of digesta with limited mixing, whereas the anatomy of the stomach and the proximal colon suggested more extensive mixing of digesta in these regions. The broad pulse of marker excretion in the wombats (fig. 1) indicated mixing in a predominantly tubular

flow of digesta. Fluctuations in marker concentration both before and after this major pulse indicated that mixing was incomplete, which resulted in both earlier excretion and prolonged retention of small amounts of digesta. Thus, digesta flow through the stomach and proximal colon are represented in figure 3 as partial mixing. Since only 4%–9% of the total digestive tract contents are attributed to the stomach (Barboza and Hume 1992*a*), small capacity may limit mixing of gastric contents. In the more capacious proximal colon, contractions of the haustra would mix digesta in several overlapping pools while peristaltic and antiperistaltic contractions of the hindgut (Stevens 1988) would intermix adjacent pools of digesta. Therefore, digesta flow through the proximal colon of the wombat is represented in figure 3 as a combination of axial flow and mixing in a series of irregular pools. Hume and Sakaguchi (1991) described this pattern of digesta flow in terms of chemical reactor theory (Penry and Jumars 1987) as modified plug flow.

TABLE 3
Intake and digestion in wombats fed unpelleted and pelleted chopped alfalfa hay

Measure and Species	Unpelleted Hay	Pelleted Hay	Statistic
Body mass (kg):			
<i>Vombatus</i>	27.2 ± 2.7	27.8 ± 2.9	NS
<i>Lasiorbinius</i>	27.5 ± 1.2	27.2 ± 1.6	
DM intake (g · kg ^{-0.75} · d ⁻¹):			
<i>Vombatus</i>	17.8 ± 1.2	29.2 ± 6.0	S**, D*
<i>Lasiorbinius</i>	38.7 ± 4.9	39.6 ± 6.1	I*
DM digestibility (%):			
<i>Vombatus</i>	68.0 ± 4.4	62.9 ± 2.6	S*, I*
<i>Lasiorbinius</i>	59.5 ± 2.0	60.7 ± 1.0	
NDF digestibility (%):			
<i>Vombatus</i>	52.1 ± 3.2	40.1 ± 8.5	S*, I*
<i>Lasiorbinius</i>	41.6 ± 2.7	38.0 ± 2.4	

Note. Values are means ± SD for four animals of each species. Statistics are as follows: S, effect of species; D, effect of diet; I, effect of species-by-diet interaction. Significance is for the *F* statistic; NS, not significant.
* *P* < 0.05.
** *P* < 0.01.

The clear separation between the MRT of fluid and particulate markers in the wombats may be attributed to mixing in the stomach and the proximal colon. Particulate digesta markers are generally retained longer than fluid markers in the stomachs of several species (Stevens 1988). In the proximal colon, contractions of the haustra would express the fluid from the coarse particulate digesta and therefore increase the rate of passage of fluid relative to the particles (Dellow 1982). Fine particles of low density may be extruded with the fluid phase and thus pass more rapidly than the coarse material.

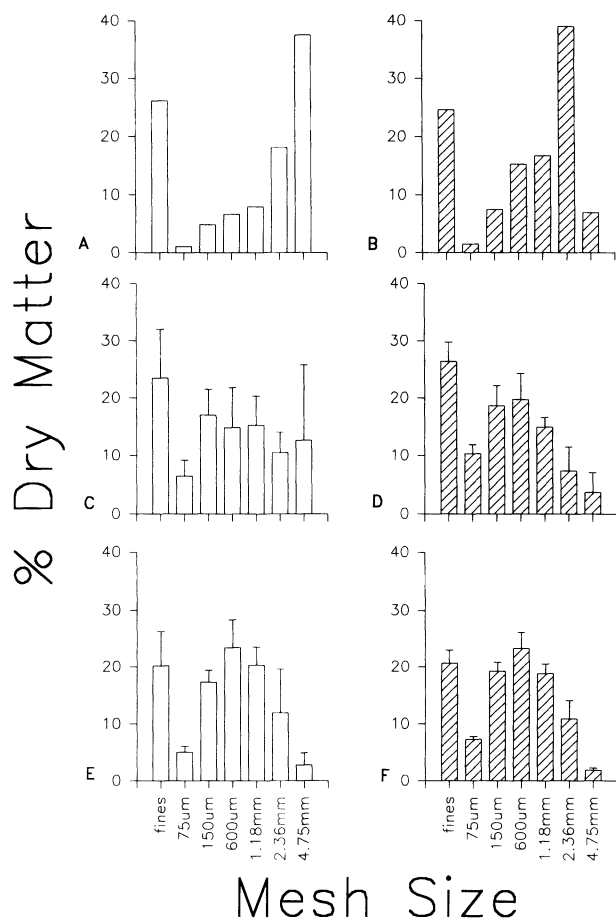


Fig. 2. Particle size distributions in unpelleted (A; open bars) and pelleted (B; hatched bars) chopped alfalfa hay and the resultant feces from *Vombatus* (C and D) and *Lasiorhinus* (E and F), expressed as DM retained (%) on 4.75-mm, 2.36-mm, 1.18-mm, 600- μ m, 150- μ m, and 75- μ m meshes. Residue passing through the smallest mesh (fines) was determined by difference. Values are means $\pm$ SD for feces from four animals of each species.

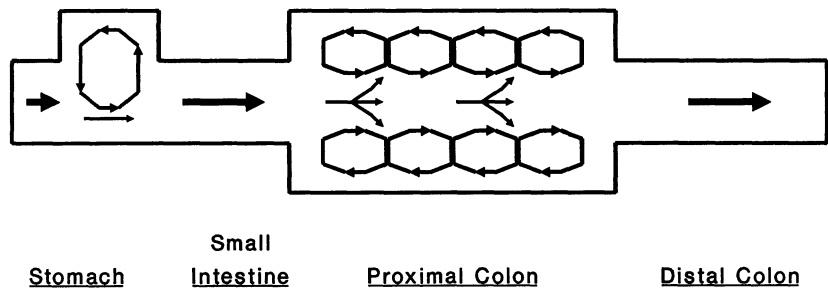


Fig. 3. Schematic model of digesta flow in wombats. Flow through the small intestine and distal colon are represented as a tubular flow with limited mixing. Partial mixing is indicated in the stomach and the proximal colon. Flow through the proximal colon is represented as a series of overlapping pools.

Sakaguchi and Hume (1990) demonstrated that fine particles moved with fluid rather than with coarse particles in two Australian possums. As more RuP was bound to fine particles than to coarse particles in vitro, retention of the coarse digesta was probably underestimated in the wombats. The separation between digesta markers in the wombats would be ameliorated by selective retention of fluid and fine particles in the distal regions of the proximal colon. Although a separation mechanism has been described in the dorsal colon of equids (Sellers et al. 1982; Bjornhag, Sperber, and Holtenius 1984), there is no direct evidence of a similar mechanism in the wombats.

The proportional length and capacity of the proximal colon are greater in *Vombatus* than in *Lasiorbini* (Barboza and Hume 1992a). The higher fiber digestibility of the low-fiber diet in *Vombatus* than in *Lasiorbini* may reflect longer retention of digesta in a larger proximal colon. Although total gut capacity is also greater in *Vombatus*, *Lasiorbini* has a longer digestive tract with a proportionately longer distal colon than *Vombatus* (Barboza and Hume 1992a). The similar retention times between the species suggest that both capacity and length of the digestive tract segments contribute to total digesta retention times in the wombats. However, this suggestion requires confirmation from direct measures of segmental digesta flow in *Vombatus* and *Lasiorbini*.

The similar sequences of DM and NDF digestion along the digestive tract of *Vombatus* and *Lasiorbini* (Barboza and Hume 1992a) suggest that differences in fecal particle size distributions between the species were related to mastication rather than digestion of the diets. Although the greater intake

of the pelleted than the unpelleted alfalfa hay in *Vombatus* was partly due to compression of the diet (Faichney 1986; Fadlalla, Kay, and Goodall 1987), the different patterns of particle size distribution in the feces of *Vombatus* suggest that dietary particle size also influenced intake (fig. 2). The smaller maximal particle size of the pelleted diet would have required less chewing (Faichney 1986), resulting in less variable fecal particle size distributions in both species. *Lasiorbhinus* chewed both unpelleted and pelleted hays to similar distributions of fecal particles, indicating a more effective dental mill than *Vombatus*. This thorough comminution of plant cell walls would enhance overall digestibility of both soluble and fibrous dietary components in *Lasiorbhinus*.

Digesta passage in the wombats was much slower than in analogous eutherian hindgut fermenters such as the equids. Retention times of RuP and CrEDTA in the wombats were longer than in horses fed chopped oat (*Avena* spp.) hay even though their body size and digestive tract capacity are smaller than those of the equids (Orton, Hume, and Leng 1985). Prolonged retention in the wombats resulted in NDF digestibilities for alfalfa hay that are similar to those of zebras (52%–45%) and tapirs (29%–39%) (Foose 1982). Neutral detergent fiber digestibilities of the high-fiber pelleted diet in the wombats were also similar to those of donkeys (31%) fed grass hay (*Vicia dasycarpa*) of similar fiber content (60.5% NDF) (Butterworth, Mosi, and Nuwanyakpa 1987). Digestible DM intakes of these donkeys were three times greater than those of the wombats on a mass-specific basis (46 vs. 14 g DM · kg^{-0.75} · d⁻¹). The low intakes of the wombats are directly related to their low metabolic requirements (Barboza et al. 1993). These low mass-specific requirements facilitate longer digesta retention and thus more extensive fiber digestion in the wombats than would be predicted from their body size (Parra 1978; Dement and Van Soest 1985).

The metabolic requirements of the wombats are also lower than those of the kangaroos and wallabies (family Macropodidae), which include several species sympatric with *Vombatus* and *Lasiorbhinus*. Unlike the wombats, the grazing macropods are foregut fermenters with well-developed tubiform forestomachs (Dellow et al. 1988; Hume 1989). Recent studies of two of these grazing macropods (*Macropus robustus erubescens* and *Macropus robustus robustus*) by Freudenberger and Hume (1992) used pelleted straw-based diets (54% and 65% NDF, 1.4% N) similar to the high-fiber diet fed to wombats. In comparison with these macropods, wombats had lower DM intakes (64–74 vs. 33 g · kg^{-0.75} · d⁻¹), longer mean retention times (27–29 vs. 52–62 h for RuP; 19–21 vs. 30–

36 h for CrEDTA), but similar digestibilities of DM (37%–45% vs. 41%–43%) and NDF (29%–33% vs. 27%–30%). Freudenberger and Hume (1992) also fed goats (*Capra hircus*) the same diets as these macropods. In comparison with the wombats, these ruminants also had higher DM intakes ($73\text{--}79 \text{ g} \cdot \text{kg}^{-0.75} \cdot \text{d}^{-1}$), shorter mean retention times (32–34 h for RuP; 27–29 h for CrEDTA), and similar digestibilities of DM (45%–53%) and NDF (27–29%).

Therefore, two contrasting digestive strategies yield similar fiber digestibilities in the grazing marsupials, slow passage through a hindgut fermentation in the wombats and faster passage through a foregut fermentation in the macropods. Foregut fermentation may utilize both the cell walls and cell contents of plants, resulting in some loss of energy as heat of fermentation of the cell contents. In the wombats, these soluble substrates are completely digested in the stomach and small intestine whereas fibrous residues are the main substrate for fermentation in the hindgut (Barboza and Hume 1992*a*, 1992*b*). This would suggest that diets high in cell contents would be more efficiently utilized by the wombats. However, a small increase in feed intake by the macropods would compensate for this small inefficiency of foregut fermentation on most forage diets. As DM intakes in the macropods are much greater than in the wombats, pasture abundance probably discriminates between these two strategies and is consistent with home ranges that are generally larger for grazing macropods (Norbury, Sanson, and Lee 1989) than for wombats.

In the arid zone, the low feed intakes and longer digesta retention times of the wombats would facilitate utilization of pastures that are often scarce and are consistent with small home ranges and foraging territories in *Lasiorhinus* (Wells 1978). Where pastures are abundant but low in quality, as is often the case in mesic regions, the low digestible intakes required by the wombats would minimize time and energy spent foraging on the same plants as macropods, or enable the utilization of even lower-quality feeds such as the sedges consumed by *Vombatus* (Barboza and Hume 1992*a*). The large digestive capacity of *Vombatus* may facilitate the utilization of abundant poor-quality vegetation, whereas a long digestive tract and a thorough dental mill may enable *Lasiorhinus* to more efficiently utilize plants that are both low in quality and quantity. These digestive strategies enable a conservative pattern of habitat use (Johnson 1991) that may allow populations of wombats to persist in areas where droughts and land degradation limit the viability of macropods and introduced herbivores.

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Literature Cited

- ALLEN, M. S., J. B. ROBERTSON, and P. J. VAN SOEST. 1984. A comparison of particle size methodologies and statistical treatments. Pages 39–56 in P. M. KENNEDY, ed. *Techniques in particle size analysis of feed and digesta in ruminants*. Canadian Society of Animal Science, Alberta.
- BAKER, D. L., and N. T. HOBBS. 1987. Strategies of digestion: digestive efficiency and retention time of forage diets in montane ungulates. *Can. J. Zool.* 65:1978–1984.
- BARBOZA, P. S. 1989. The nutritional physiology of the Vombatidae. Ph.D. thesis. University of New England, Armidale, N.S.W., Australia. 162 pp.
- BARBOZA, P. S., and I. D. HUME. 1992*a*. Digestive tract morphology and digestion in the wombats (Marsupialia: Vombatidae). *J. Comp. Physiol.* 162B:552 – 560.
- . 1992*b*. Hindgut fermentation in the wombats: two marsupial grazers. *J. Comp. Physiol.* 162B:561 – 566.
- BARBOZA, P. S., I. D. HUME, and J. V. NOLAN. 1993. Nitrogen metabolism and requirements of nitrogen and energy in the wombats (Marsupialia: Vombatidae). *Physiol. Zool.* 66:807–828.
- BJORNHAG, G., I. SPERBER, and K. HOLTENIUS. 1984. A separation mechanism in the large intestine of equines. *Can. J. Anim. Sci.* 64(suppl.):89–90.
- BLAXTER, K. L., N. MCC. GRAHAM, and F. W. WAINMAN. 1956. Some observations on the digestibility of food by sheep, and on related problems. *Br. J. Nutr.* 10:69–91.
- BUTTERWORTH, M. H., A. K. MOSI, and M. NUWANYAKPA. 1987. Intake and digestibility of roughages by donkeys. Pages 105–106 in M. ROSE, ed. *Herbivore nutrition research*. Australian Society of Animal Production, Brisbane.
- DELLOW, D. W. 1982. Studies on the nutrition of macropodine marsupials. III. The flow of digesta through the stomach and intestine of macropodines and sheep. *Aust. J. Zool.* 30:751–765.
- DELLOW, D. W., I. D. HUME, R. T. J. CLARKE, and T. BAUCHOP. 1988. Microbial activity in the forestomach of free-living macropodid marsupials: comparisons with laboratory studies. *Aust. J. Zool.* 36:383–395.

- DEMMENT, M. W., and P. J. VAN SOEST. 1985. A nutritional explanation for body-size patterns of ruminant and non-ruminant herbivores. *Am. Nat.* 125:641–672.
- DIXON, R. M., J. J. KENNELLY, and L. P. MILLIGAN. 1983. Kinetics of [^{103}Ru]phenanthroline and dysprosium particulate markers in the rumen of steers. *Br. J. Nutr.* 49:463–473.
- DOWNES, A. M., and I. W. McDONALD. 1964. The chromium-51 complex of ethylene-diamine tetraacetic acid as a soluble rumen marker. *Br. J. Nutr.* 18:153–162.
- EVANS, E. W., G. R. PEARCE, J. BURNETT, and S. L. PILLINGER. 1973. Changes in some physical characteristics of the digesta in the reticulo-rumen of cows fed once daily. *Br. J. Nutr.* 29:357–376.
- FADLALLA, B., R. N. B. KAY, and E. D. GOODALL. 1987. Effects of particle size on digestion of hay by sheep. *J. Agric. Sci.* 109:551–561.
- FAICHNEY, G. J. 1986. The kinetics of particulate matter in the rumen. Pages 173–195 in L. P. MILLIGAN, W. L. GROVUM, and A. DOBSON, eds. *Control of digestion and metabolism in ruminants*. Prentice-Hall, Englewood Cliffs, N.J.
- FOOSE, T. J. 1982. Trophic strategies of ruminant versus nonruminant ungulates. Ph.D. thesis. University of Chicago.
- FREUDENBERGER, D. O., and I. D. HUME. 1992. Ingestive and digestive responses to dietary fibre and nitrogen by two macropodid marsupials (*Macropus robustus erubescens* and *M. r. robustus*) and a ruminant (*Capra hircus*). *Aust. J. Zool.* 40:191–194.
- GAUGHWIN, M. D. 1981. Socioecology of the hairy-nosed wombat (*Lasiorhinus latifrons*) in the Blanchetown region of South Australia. Ph.D. thesis. University of Adelaide. 137 pp.
- HUME, I. D. 1989. Optimal digestive strategies in mammalian herbivores. *Physiol. Zool.* 62:1145–1163.
- HUME, I. D., and P. S. BARBOZA. 1993. Designing artificial diets for captive marsupials. Pages 281–288 in M. E. FOWLER, ed. *Zoo and wild animal medicine*. 3d ed. Saunders, Orlando, Fla.
- HUME, I. D., and E. SAKAGUCHI. 1991. Patterns of digesta flow and digestion in foregut and hindgut fermenters. Pages 427–451 in T. TSUDA, Y. SASAKI, and J. KAWASHIMA, eds. *Physiological aspects of digestion and metabolism in the ruminants*. Academic Press, San Diego, Calif.
- ILLIUS, A. W., and I. J. GORDON. 1992. Modeling the nutritional ecology of ungulate herbivores: evolution of body size and competitive interactions. *Oecologia* 89:428–434.
- JOHNSON, C. N. 1991. Utilization of habitat by the northern hairy-nosed wombat *Lasiorhinus kreftii*. *J. Zool. Lond.* 225:495–507.
- McILROY, J. C. 1973. Aspects of the ecology of the common wombat *Vombatus ursinus* (Shaw 1800). Ph.D. thesis. Australian National University. 250 pp.
- . 1976. Aspects of the ecology of the common wombat *Vombatus ursinus*. I. Capture, handling, marking and radiotracking techniques. *Aust. Wildl. Res.* 3:105–116.
- NORBURY, G. L., G. D. SANSON, and A. K. LEE. 1989. Feeding ecology of the Macropodoidea. Pages 169–178 in G. GRIGG, P. JARMAN, and I. HUME, eds. *Kangaroos, wallabies and rat-kangaroos*. Surrey Beatty, Chipping Norton.

- ORTON, R. K., I. D. HUME, and R. A. LENG. 1985. Effects of exercise and level of dietary protein on digestive function in horses. *Equine Vet. J.* 17:386–390.
- OWENS, F. N., and C. F. HANSON. 1992. External and internal markers for appraising site and extent of digestion in ruminants. *J. Dairy Sci.* 75:2605–2617.
- PARRA, R. 1978. Comparison of foregut and hindgut fermentation in herbivores. Pages 205–229 in G. G. MONTGOMERY, ed. *The ecology of arboreal folivores*. Smithsonian Institution Press, Washington, D.C.
- PENRY, D. L., and P. A. JUMARS. 1987. Modeling animal guts as chemical reactors. *Am. Nat.* 129:69–96.
- SAKAGUCHI, E., and I. D. HUME. 1990. Digesta retention and fibre digestion in brushtail possums, ringtail possums and rabbits. *Comp. Biochem. Physiol.* 96A:351–354.
- SELLERS, A. F., J. E. LOWE, C. J. DROST, V. T. RENDANO, J. R. GEORGI, and M. C. ROBERTS. 1982. Retropulsion-propulsion in equine large colon. *Am. J. Vet. Res.* 43:390–396.
- SPALINGER, D. E., and N. T. HOBBS. 1992. Mechanisms of foraging in mammalian herbivores: new models of functional response. *Am. Nat.* 140:325–348.
- STEVENS, C. E. 1988. *Comparative physiology of the vertebrate digestive system*. Cambridge University Press, New York. 300 pp.
- TABACHNICK, B. G., and L. S. FIDELL. 1983. *Using multivariate statistics*. Harper & Row, New York. 509 pp.
- TAN, T. N., R. H. WESTON, and J. P. HOGAN. 1971. Use of ¹⁰³Ru-labeled tris (1,10-phenanthroline) Ruthenium (II) Chloride as a marker in digestion studies with sheep. *Int. J. Appl. Radiat. Isot.* 22:301–308.
- VAN SOEST, P. J., J. B. ROBERTSON, and B. A. LEWIS. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74:3583–3597.
- WARNER, A. C. I. 1981. Rate of passage of digesta through the gut of mammals and birds. *Nutr. Abstr. Rev.* 51:789–820.
- WELLS, R. T. 1978. Field observations of the hairy-nosed wombat, *Lasiorhinus latifrons* (Owen). *Aust. Wildl. Res.* 5:299–303.
- WINER, B. J. 1971. *Statistical principles in experimental design*. 2d ed. McGraw-Hill, New York. 907 pp.
- ZAR, J. H. 1974. *Biostatistical analysis*. Prentice-Hall, NSW, Australia. 620 pp.